

Uniqueness

HW8, problem 6

$$A = \begin{bmatrix} 1+i & 2+i & 1+2i \\ 1-i & 2-i & 1-2i \\ 2 & 5 & 5 \end{bmatrix}$$

$$\mathbb{C}[i]^3 / \text{im } A \leftarrow \text{What is this?}$$

$$1 = (-i) \cdot (1+i) + 1 \cdot (2+i)$$

$$\xrightarrow{C_1 \leftrightarrow C_2} \begin{bmatrix} 2+i & 1+i & 1+2i \\ 2-i & 1-i & 1-2i \\ 5 & 2 & 5 \end{bmatrix}$$

$$\underline{C1 - C2} \rightarrow$$

$$\begin{bmatrix} 1 & 1+i & 1+2i \\ 1 & 1-i & 1-2i \\ 3 & 2 & 5 \end{bmatrix}$$

$$\underline{R2 - R1}$$

$$R3 - 3 \cdot R1$$

$$\begin{bmatrix} 1 & 1+i & 1+2i \\ 0 & -2i & -4i \\ 0 & -1-3i & 2-6i \end{bmatrix}$$

$$\underline{C2 - (1+i) \cdot C1}$$

$$C3 - (1+2i) \cdot C1$$

$$C2 \leftrightarrow C3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2i & -4i \\ 0 & 1+3i & 2-6i \end{bmatrix}$$

$$1+3i = 2i + (1+i)$$

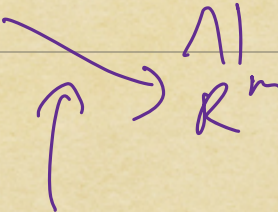
$$2i = (1+i)(1+i)$$

State of things:

N : f.s. R -module.

$$(1) \quad R^m / \mathfrak{m} \xrightarrow{\cong} N \quad \text{for some } m \geq r$$

$$(2) \quad R^n \xrightarrow{\cong} \mathfrak{m}, \quad n \leq m$$



$(m \times n)$ -matrix A

$$(3) \quad N \cong R^m / \text{im}(A)$$

$$\cong R^m / \text{im}(\text{SNF}(A))$$

$$(4) \quad \text{SNF}(A) = \begin{bmatrix} r_1 & & & 0 \\ & r_2 & & \\ & & \ddots & \\ & & & r_r & & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & & & \\ & \ddots & & \\ & & 0 & \\ & & & \ddots \end{bmatrix}$$

$$r_1 | r_2 | \dots | r_t$$

$$R^m / \text{im}(\text{SNF}(A))$$

$$\cong R(r_1) \times \dots \times R(r_t) \times R^{\boxed{m-t}}$$

We have shown the
existence part of the
fundamental theorem.

Uniqueness of rank

Defⁿ $N: R$ -module
 ψ
 n

(a) N is torsion if $\exists r \in R \setminus \{0\}$
s.t. $r \cdot n = 0$

(b) N is torsion-free if
it has no non-zero torsion elements

Example: (i) Because R is
an integral domain, R is torsion-free
as an R -module

$\Rightarrow R^m$ is torsion-free

(2) $r \in R \setminus \{0\}$, r : non-invertible

$N = R/(r) \ni a + (r)$ is torsion
since

$$\begin{array}{ccc} r \cdot (a + (r)) & = & (r) \\ \parallel & & \parallel \\ r \cdot a + (r) & & \end{array}$$

Reh. If r is irreducible, then $R/(r)$ is a field, and as a module over itself it is torsion-free. But as a module over R it is all torsion

(3) $R = \mathbb{F}_5[x]$

$\mathbb{F}_5[x]/(x^2-2)$: field but torsion as an $\mathbb{F}_5[x]$ -module

Defⁿ N : R -module

$$(c) \quad N_{\text{tors}} = \{n \in N : n \text{ is torsion}\}$$

Observation

$N_{\text{tors}} \leq N$ is an R -submodule

Pf:

$$\bullet \quad r_1 n_1 = 0, \quad r_2 n_2 = 0$$

$$\text{then } r_1 r_2 (n_1 + n_2) = 0$$

$$[r_1 \neq 0, r_2 \neq 0 \Rightarrow r_1 r_2 \neq 0]$$

$$\bullet \quad r \cdot n = 0, \quad s \in R$$

$$r \cdot (s n) = s \cdot (r n) = s \cdot 0 = 0$$

$$(d) \quad N_{\text{tf}} = N / N_{\text{tors}}$$

Observation:

N_{tf} is torsion-free

$$\uparrow \uparrow: \quad n \in N_{tf} \quad , \quad r \in R \text{ s.t. } r \cdot n = 0$$

$$\parallel$$
$$m \in N_{tors}$$

$$\downarrow$$

$$r \cdot m \in N_{tors} = 0$$

$$\Leftrightarrow r \cdot m \in N_{tors}$$

$$\Rightarrow \exists s \in R \setminus \{0\}$$

$$\text{s.t. } s \cdot (r \cdot m) = 0$$

$$\parallel$$
$$(sr) \cdot m$$

$$\Leftrightarrow m \in N_{tors}$$

$$\Rightarrow n = 0 \in N_{tf}.$$

We find:

$$N_{\text{tors}} \leq N$$

all torsion

$$N/N_{\text{tors}} = N_{\text{tf.}}$$

torsion-free

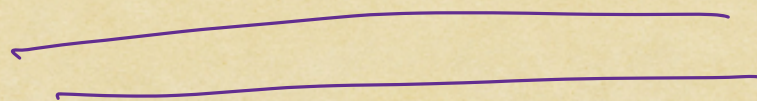
Observation

$$r_1, \dots, r_t \in R \setminus \{0\}$$

$$N = R/(r_1) \times \dots \times R/(r_t) \times R^s$$

$$N_{\text{tors}} = R/(r_1) \times \dots \times R/(r_t) \times \{0\}$$

N_{tf}



R^s

Proposition: The rank s is
determined uniquely by N

Pf: The precise statement :

$$\text{If } N \cong R/(s'_1) \times \cdots \times R/(s'_n) \times R^{s'}$$

then $s = s'$

Need to show:

$$R^s \cong R^{s'} \Leftrightarrow$$

$$s = s'$$

$$R^S \xrightarrow[\uparrow]{\cong} R^{S'}$$

$S' \times S$ -matrix A

Can assume $A = \begin{bmatrix} r_1 & & & \\ & \ddots & & \\ & & r_{t-1} & \\ & & & 0 \end{bmatrix}$

$\underbrace{\hspace{10em}}_S$

$\underbrace{\hspace{2em}}_{S'}$

This gives an isom.

$(\Leftarrow)$

- $S = S'$

- Every diagonal entry is invertible